

6663/01

Edexcel GCE

Core Mathematics C1

Advanced Subsidiary

Coordinate Geometry: Straight Lines

Calculators may NOT be used for these questions.

Information for Candidates

A booklet 'Mathematical Formulae and Statistical Tables' might be needed for some questions.

The marks for the parts of questions are shown in round brackets, e.g. **(2)**.

There are 23 questions in this test.

Advice to Candidates

You must ensure that your answers to parts of questions are clearly labelled.

You must show sufficient working to make your methods clear.

Answers without working may not gain full credit.

1. (a) Find an equation of the line joining $A(7, 4)$ and $B(2, 0)$, giving your answer in the form $ax + by + c = 0$, where a , b and c are integers. (3)

- (b) Find the length of AB , leaving your answer in surd form. (2)

The point C has coordinates $(2, t)$, where $t > 0$, and $AC = AB$.

- (c) Find the value of t . (1)

- (d) Find the area of triangle ABC . (2)
- (Total 8 marks)**
-

2. The curve C has equation $y = f(x)$, $x > 0$, where

$$\frac{dy}{dx} = 3x - \frac{5}{\sqrt{x}} - 2$$

Given that the point $P(4, 5)$ lies on C , find

- (a) $f(x)$, (5)

- (b) an equation of the tangent to C at the point P , giving your answer in the form $ax + by + c = 0$, where a , b and c are integers. (4)
- (Total 9 marks)**
-

3. The line l_1 has equation $3x + 5y - 2 = 0$

(a) Find the gradient of l_1 .

(2)

The line l_2 is perpendicular to l_1 and passes through the point $(3, 1)$.

(b) Find the equation of l_2 in the form $y = mx + c$, where m and c are constants.

(3)

(Total 5 marks)

4. (a) Factorise completely $x^3 - 4x$

(3)

(b) Sketch the curve C with equation

$$y = x^3 - 4x,$$

showing the coordinates of the points at which the curve meets the x -axis.

(3)

The point A with x -coordinate -1 and the point B with x -coordinate 3 lie on the curve C .

(c) Find an equation of the line which passes through A and B , giving your answer in the form $y = mx + c$, where m and c are constants.

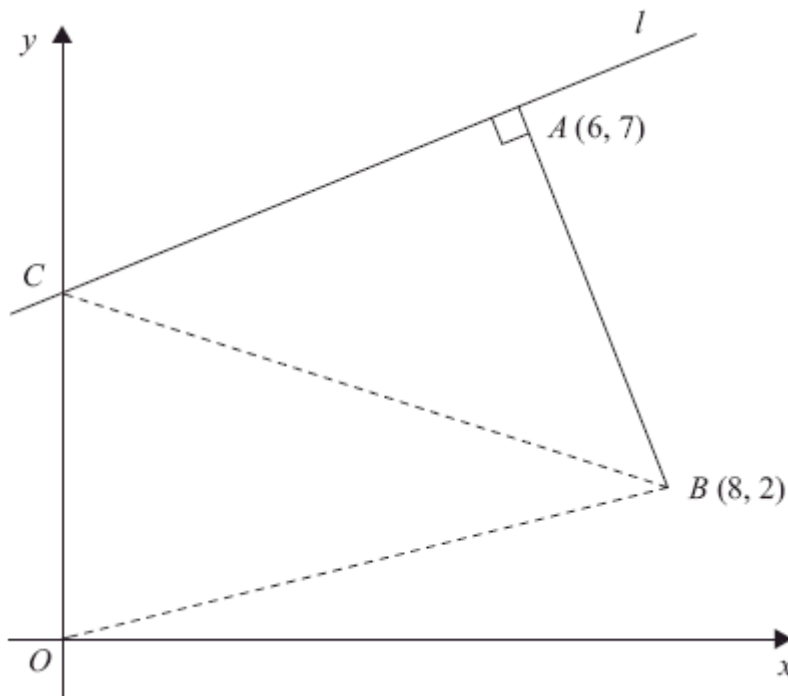
(5)

(d) Show that the length of AB is $k\sqrt{10}$, where k is a constant to be found.

(2)

(Total 13 marks)

5.



The points A and B have coordinates $(6, 7)$ and $(8, 2)$ respectively.

The line l passes through the point A and is perpendicular to the line AB , as shown in the diagram above.

- (a) Find an equation for l in the form $ax + by + c = 0$, where a , b and c are integers.

(4)

Given that l intersects the y -axis at the point C , find

- (b) the coordinates of C ,

(2)

- (c) the area of $\triangle OCB$, where O is the origin.

(2)

(Total 8 marks)

6. The curve C has equation

$$y = x^3 - 2x^2 - x + 9, \quad x > 0$$

The point P has coordinates $(2, 7)$.

- (a) Show that P lies on C .

(1)

- (b) Find the equation of the tangent to C at P , giving your answer in the form $y = mx + c$, where m and c are constants.

(5)

The point Q also lies on C .

Given that the tangent to C at Q is perpendicular to the tangent to C at P ,

- (c) show that the x -coordinate of Q is $\frac{1}{3}(2 + \sqrt{6})$.

(5)

(Total 11 marks)

7. The line l_1 passes through the point $A(2, 5)$ and has gradient $-\frac{1}{2}$.

(a) Find an equation of l_1 , giving your answer in the form $y = mx + c$.

(3)

The point B has coordinates $(-2, 7)$.

(b) Show that B lies on l_1 .

(1)

(c) Find the length of AB , giving your answer in the form $k\sqrt{5}$, where k is an integer.

(3)

The point C lies on l_1 and has x -coordinate equal to p .

The length of AC is 5 units.

(d) Show that p satisfies

$$p^2 - 4p - 16 = 0.$$

(4)

(Total 11 marks)

8. The curve C has equation

$$y = 9 - 4x - \frac{8}{x}, \quad x > 0.$$

The point P on C has x -coordinate equal to 2.

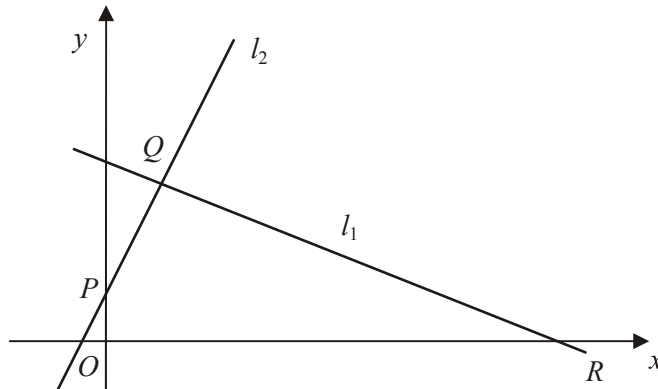
- (a) Show that the equation of the tangent to C at the point P is $y = 1 - 2x$. (6)

- (b) Find an equation of the normal to C at the point P . (3)

The tangent at P meets the x -axis at A and the normal at P meets the x -axis at B .

- (c) Find the area of triangle APB . (4)
- (Total 13 marks)**
-

9.



The points $Q(1, 3)$ and $R(7, 0)$ lie on the line l_1 , as shown in the diagram above.

The length of QR is $a\sqrt{5}$.

(a) Find the value of a .

(3)

The line l_2 is perpendicular to l_1 , passes through Q and crosses the y -axis at the point P , as shown in the diagram above.

Find

(b) an equation for l_2 ,

(5)

(c) the coordinates of P ,

(1)

(d) the area of $\triangle PQR$.

(4)

(Total 13 marks)

10. The point $A(-6, 4)$ and the point $B(8, -3)$ lie on the line L .

(a) Find an equation for L in the form $ax + by + c = 0$, where a , b and c are integers.

(4)

(b) Find the distance AB , giving your answer in the form $k\sqrt{5}$, where k is an integer.

(3)

(Total 7 marks)

11. The line l_1 has equation $y = 3x + 2$ and the line l_2 has equation $3x + 2y - 8 = 0$.

(a) Find the gradient of the line l_2 .

(2)

The point of intersection of l_1 and l_2 is P .

(b) Find the coordinates of P .

(3)

The lines l_1 and l_2 cross the line $y = 1$ at the points A and B respectively.

(c) Find the area of triangle ABP .

(4)

(Total 9 marks)

12. The line l_1 passes through the points $P(-1, 2)$ and $Q(11, 8)$.

- (a) Find an equation for l_1 in the form $y = mx + c$, where m and c are constants.

(4)

The line l_2 passes through the point $R(10, 0)$ and is perpendicular to l_1 . The lines l_1 and l_2 intersect at the point S .

- (b) Calculate the coordinates of S .

(5)

- (c) Show that the length of RS is $3\sqrt{5}$.

(2)

- (d) Hence, or otherwise, find the exact area of triangle PQR .

(4)

(Total 15 marks)

13. The line L has equation $y = 5 - 2x$.

- (a) Show that the point $P(3, -1)$ lies on L .

(1)

- (b) Find an equation of the line perpendicular to L , which passes through P . Give your answer in the form $ax + by + c = 0$, where a , b and c are integers.

(4)

(Total 5 marks)

14. The line l_1 passes through the point $(9, -4)$ and has gradient $\frac{1}{3}$.

(a) Find an equation for l_1 in the form $ax + by + c = 0$, where a , b and c are integers.

(3)

The line l_2 passes through the origin O and has gradient -2 . The lines l_1 and l_2 intersect at the point P .

(b) Calculate the coordinates of P .

(4)

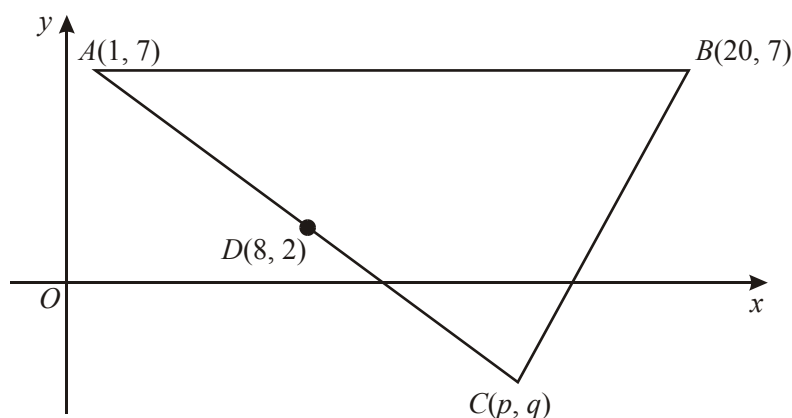
Given that l_1 crosses the y -axis at the point C ,

(c) calculate the exact area of $\triangle OCP$.

(3)

(Total 10 marks)

15.



The points $A(1, 7)$, $B(20, 7)$ and $C(p, q)$ form the vertices of a triangle ABC , as shown in the diagram. The point $D(8, 2)$ is the mid-point of AC .

(a) Find the value of p and the value of q .

(2)

The line l , which passes through D and is perpendicular to AC , intersects AB at E .

(b) Find an equation for l , in the form $ax + by + c = 0$, where a , b and c are integers.

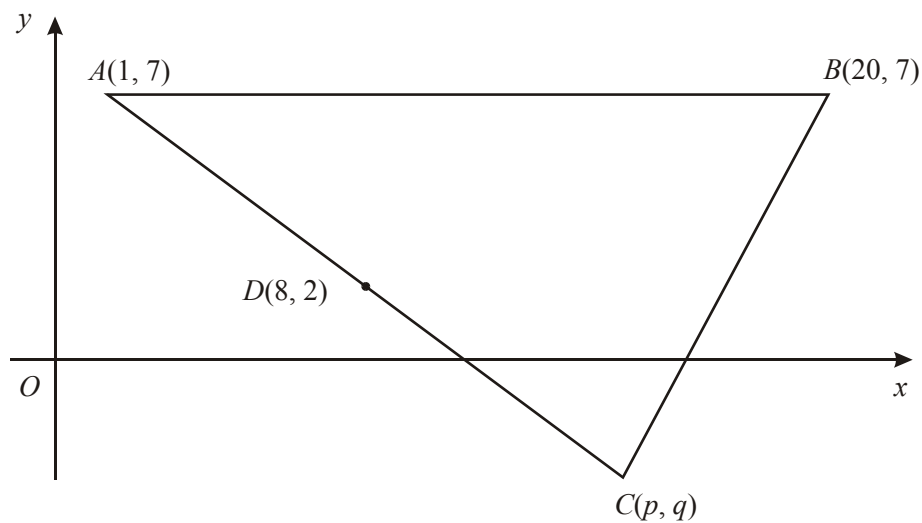
(5)

(c) Find the exact x -coordinate of E .

(2)

(Total 9 marks)

16.



The points $A(1, 7)$, $B(20, 7)$ and $C(p, q)$ form the vertices of a triangle ABC , as shown in the diagram. The point $D(8, 2)$ is the mid-point of AC .

- (a) Find the value of p and the value of q .

(2)

The line l , which passes through D and is perpendicular to AC , intersects AB at E .

- (b) Find an equation for l , in the form $ax + by + c = 0$, where a , b and c are integers.

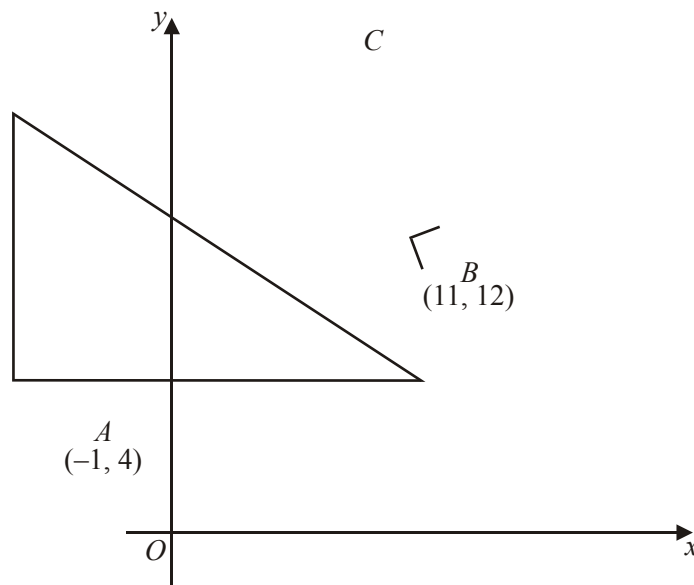
(5)

- (c) Find the exact x -coordinate of E .

(2)

(Total 9 marks)

17.



In $\triangle ABC$, the coordinates of A and B are $(-1, 4)$ and $(11, 12)$ respectively and $\angle ABC = 90^\circ$, as shown in the diagram above.

- (a) Find the gradient of AB .

(2)

The line l passes through the points B and C .

- (b) Find an equation of l in the form $px + qy + r = 0$, where p , q and r are integers to be found.

(4)

The line l crosses the x -axis at the point D . Given that B is the mid-point of CD ,

- (c) find the coordinates of C ,

(3)

- (d) show that $AC = 4\sqrt{26}$.

(2)

(Total 11 marks)

18. The points A and B have coordinates $(1, 2)$ and $(5, 8)$ respectively.

(a) Find the coordinates of the mid-point of AB .

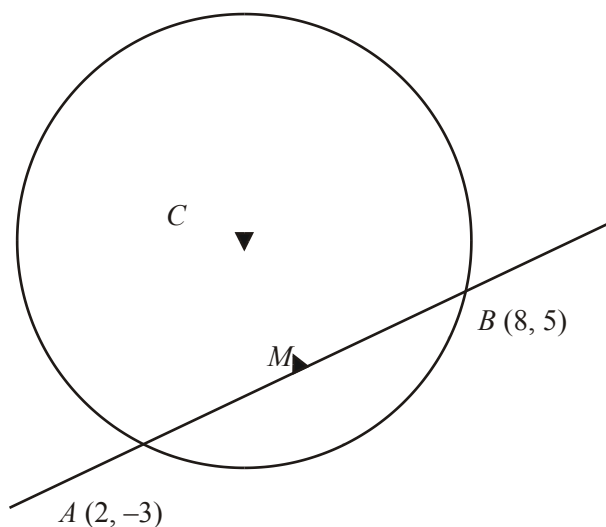
(2)

(b) Find, in the form $y = mx + c$, an equation for the straight line through A and B .

(4)

(Total 6 marks)

19.



The points A and B have coordinates $(2, -3)$ and $(8, 5)$ respectively, and AB is a chord of a circle with centre C , as shown in the diagram above.

(a) Find the gradient of AB .

(2)

The point M is the mid-point of AB .

(b) Find an equation for the line through C and M .

(5)

Given that the x -coordinate of C is 4,

(c) find the y -coordinate of C ,

(2)

(d) show that the radius of the circle is $\frac{5\sqrt{17}}{4}$.

(4)

Total 13 marks

- 20.** The points A and B have coordinates $(4, 6)$ and $(12, 2)$ respectively.

The straight line l_1 passes through A and B .

- (a) Find an equation for l_1 in the form $ax + by = c$, where a , b and c are integers.

(4)

The straight line l_2 passes through the origin and has gradient -4 .

- (b) Write down an equation for l_2 .

(1)

The lines l_1 and l_2 intersect at the point C .

- (c) Find the exact coordinates of the mid-point of AC .

(5)

(Total 10 marks)

- 21.** The straight line l_1 with equation $y = \frac{3}{2}x - 2$ crosses the y -axis at the point P . The point Q has coordinates $(5, -3)$.

- (a) Calculate the coordinates of the mid-point of PQ .

(3)

The straight line l_2 is perpendicular to l_1 and passes through Q .

- (b) Find an equation for l_2 in the form $ax + by = c$, where a , b and c are integer constants.

(4)

The lines l_1 and l_2 intersect at the point R .

- (c) Calculate the exact coordinates of R .

(4)

(Total 11 marks)

- 22.** (a) Find an equation of the line p which passes through the point $(-3, 2)$ and which is parallel to the line q with equation $7x - 2y - 14 = 0$.

(3)

The lines p and q meet the y -axis at the points A and B respectively.

- (b) Find the distance between AB .

(2)

(Total 5 marks)

23. The straight line l_1 has equation $y = 3x - 6$.

The straight line l_2 is perpendicular to l_1 and passes through the point $(6, 2)$.

(a) Find an equation for l_2 in the form $y = mx + c$, where m and c are constants.

(3)

The lines l_1 and l_2 intersect at the point C .

(b) Use algebra to find the coordinates of C .

(3)

The lines l_1 and l_2 cross the x -axis at the points A and B respectively.

(c) Calculate the exact area of triangle ABC .

(4)

(Total 10 marks)

1. (a) $m_{AB} = \frac{4-0}{7-2} \left(= \frac{4}{5} \right)$ M1

Equation of AB is: $y-0 = \frac{4}{5}(x-2)$ or $y-4 = \frac{4}{5}(x-7)$ (o.e.) M1

$\underline{4x-5y-8=0}$ (o.e.) A1 3

Note

Apply the usual rules for quoting formulae here.

For a correctly quoted formula with some correct substitution award M1

If no formula is quoted then a fully correct expression is needed for the M mark

1st M1 for attempt at gradient of AB. Some correct substitution in correct formula.

2nd M1 for an attempt at equation of AB. Follow through their gradient, not e.g. $-\frac{1}{m}$

Using $y = mx + c$ scores this mark when c is found.

Use of $\frac{y-y_1}{y_2-y_1} = \frac{x-x_1}{x_2-x_1}$ scores 1st M1 for denominator, 2nd M1 for

use of a correct point

A1 requires integer form but allow $5y + 8 = 4x$ etc.

Must have an “=” or A0

(b) $(AB =) \sqrt{(7-2)^2 + (4-0)^2}$ M1

$= \sqrt{41}$ A1 2

Note

M1 for an expression for AB or AB^2 . Ignore what is “left” of the equals sign

(c) Using isos triangle with $AB = AC$ then $t = 2 \times y_A = 2 \times 4 = 8$ B1 1

Note

B1 for $t = 8$. May be implied by correct coordinates (2, 8) or the value appearing in (d)

(d) Area of triangle $= \frac{1}{2}t \times (7-2)$ M1

$= \underline{20}$ A1 2

Note

M1 for an expression for the area of the triangle, follow through their t ($\neq 0$) but must have the $(7-2)$ or 5 and the $\frac{1}{2}$.

DET

e.g. $\begin{array}{cccc} 2 & 7 & 2 & 2 \\ 0 & 4 & t & 0 \end{array}$ Area = $\frac{1}{2} [8 + 7t + 0 - (0 + 8 + 2t)]$ Must have the $\frac{1}{2}$ for M1

[8]

2. (a) $(y =) \frac{3x^2}{2} - \frac{5x^{\frac{1}{2}}}{\frac{1}{2}} - 2x (+c)$

M1 A1 A1

$$f(4) = 5 \Rightarrow 5 = \frac{3}{2} \times 16 - 10 \times 2 - 8 + c$$

M1

$$\underline{c = 9}$$

A1

5

$$\left[f(x) = \frac{3}{2}x^2 - 10x^{\frac{1}{2}} - 2x + 9 \right]$$

Note

1st M1 for an attempt to integrate $x^n \rightarrow x^{n+1}$

1st A1 for at least 2 correct terms in x (unsimplified)

2nd A1 for all 3 terms in x correct (condone missing $+c$ at this point).
Needn't be simplified

2nd M1 for using the point (4, 5) to form a linear equation for c .
Must use $x = 4$ and $y = 5$ and have no x term and the function must have "changed".

3rd A1 for $c = 9$. The final expression is not required.

(b) $m = 3 \times 4 - \frac{5}{2} - 2 \left(= 7.5 \text{ or } \frac{15}{2} \right)$

M1

Equation is: $y - 5 = \frac{15}{2}(x - 4)$

M1 A1

$$\underline{2y - 15x + 50 = 0} \quad \text{o.e.}$$

A1

4

Note

1st M1 for an attempt to evaluate $f'(4)$. Some correct use of $x = 4$ in $f'(x)$ but condone slips. They must therefore have at least 3×4 or $-\frac{5}{2}$ and clearly be using $f'(x)$ with $x = 4$. Award this mark wherever it is seen

2nd M1 for using their value of m [or their $-\frac{1}{m}$ (provided it clearly comes from using $x = 4$ in $f'(x)$) to form an equation of the line through $(4, 5)$).

Allow this mark for an attempt at a normal or tangent. Their m must be numerical. Use of $y = mx + c$ scores this mark when c is found.

1st A1 for any correct expression for the equation of the line

2nd A1 for any correct equation with integer coefficients. An “=” is required. e.g. $2y = 15x - 50$ etc as long as the equation is correct and has integer coefficients.

Normal

Attempt at normal can score both M marks in (b) but A0A0

[9]

3. (a) Putting the equation in the form $y = mx (+c)$ and attempting to extract the m or mx (not the c), M1
or finding 2 points on the line and using the correct gradient formula.

Gradient = $-\frac{3}{5}$ (or equivalent) A1 2

Note

Condone sign errors and ignore the c for the M mark, so... both marks can be scored even if c is wrong (e.g. $c = -\frac{2}{5}$) or omitted.

Answer only: $-\frac{3}{5}$ scores M1 A1. Any other answer only scores M0 A0.

$y = -\frac{3}{5}x + \frac{2}{5}$ with no further progress scores

M0 A0 (m or mx not extracted).

- (b) Gradient of perp. line = $-\frac{1}{\left(-\frac{3}{5}\right)}$ (Using $-\frac{1}{m}$ with the m from

part (a)) M1

$$y - 1 = \left(\frac{5}{3}\right)(x - 3) \quad \text{M1}$$

$$y = -\frac{5}{3}x - 4 \quad (\text{Must be in this form... allow } y = \frac{5}{3}x - \frac{12}{3} \text{ but}$$

$$\text{not } y = \frac{5x - 12}{3}) \quad \text{A1} \quad 3$$

This A mark is dependent upon both M marks.

Note

2nd M: For the equation, in any form, of a straight line through (3, 1) with any numerical gradient (except 0 or ∞).

(Alternative is to use (3, 1) in $y = mx + c$ to find a value for c ,

in which case $y = \frac{5}{3}x + c$ leading to $c = -4$ is sufficient for the A1).

[5]

4. (a) $x(x^2 - 4)$ Factor x seen in a correct factorised form of the expression. B1

$$= x(x - 2)(x + 2) \quad \text{M: Attempt to factorise quadratic (general principles).} \quad \text{M1 A1} \quad 3$$

Accept $(x - 0)$ or $(x + 0)$ instead of x at any stage.

Factorisation must be seen in part (a) to score marks.

Note

$$x^3 - 4x \rightarrow x(x^2 - 4) \rightarrow (x - 2)(x + 2) \quad \text{scores B1 M1 A0.}$$

$$x^3 - 4x \rightarrow x^2 - 4 \rightarrow (x - 2)(x + 2) \quad \text{scores B0 M1 A0 (dividing by } x).$$

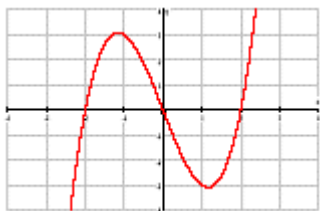
$$x^3 - 4x \rightarrow x(x^2 - 4x) \rightarrow x^2(x - 4) \quad \text{scores B0 M1 A0.}$$

$$x^3 - 4x \rightarrow x(x^2 - 4) \rightarrow x(x - 2)^2 \quad \text{scores B1 M1 A0}$$

$$\text{Special cases: } x^3 - 4x \rightarrow (x - 2)(x^2 + 2x) \quad \text{scores B0 M1 A0.}$$

$$x^3 - 4x \rightarrow x(x - 2)^2 \quad (\text{with no intermediate step seen}) \quad \text{scores B0 M1 A0}$$

(b)



Shape  (2 turning points required)

B1

Through (or touching) origin

B1

Crossing x -axis or “stopping at x -axis” (not a turning point) at $(-2, 0)$ and $(2, 0)$.

B1 3

Allow -2 and 2 on x -axis. Also allow $(0, -2)$ and $(0, 2)$ if marked on x -axis.

Ignore extra intersections with x -axis.

Note

The 2nd and 3rd B marks are not dependent upon the 1st B mark, but are dependent upon a sketch having been attempted.

(c) Either $y = 3$ (at $x = -1$) or $y = 15$ (at $x = 3$)

Allow if seen elsewhere.

B1

Gradient = $\frac{15-3}{3-(-1)} (= 3)$ Attempt correct grad. formula

with their y values.

M1

For gradient M mark, if correct formula not seen, allow one slip, e.g. $\frac{15-3}{3-1}$

$y - 15 = m(x - 3)$ or $y - 3 = m(x - (-1))$, with any value for m .

M1

$y - 15 = 3(x - 3)$ or the correct equation in any form,

A1

e.g. $y - 3 = \frac{15-3}{3-(-1)}(x - (-1))$, $\frac{y-3}{x+1} = \frac{15-3}{3+1}$

$y = 3x + 6$

A1 5

Note

1st M: May be implicit in the equation of the line, e.g.

$$\frac{y-15}{3-15} = \frac{x-3}{-1-3}$$

2nd M: An equation of a line through (3, “15”) or (–1, “3”) in any form,

with any gradient (except 0 or ∞).

2nd M: Alternative is to use one of the points in $y = mx + c$ to find a value

for c , in which case $y = 3x + c$ leading to $c = 6$ is sufficient for both A marks.

1st A1: Correct equation in any form.

(d) $AB = \sqrt{("15-3")^2 + (3-(-1))^2}$ (With their non-zero y values)...

Square root is required.

M1

$\sqrt{160} (= \sqrt{16}\sqrt{10}) = 4\sqrt{10}$ (Ignore $\pm$ if seen) ($\sqrt{16}\sqrt{10}$ need not be seen).

A1 2

[13]

5. (a) $AB : m = \frac{2-7}{8-6} \left(= -\frac{5}{2} \right)$

B1

Using $m_1 m_2 = -1$: $m_2 = \frac{2}{5}$

M1

$y-7 = \frac{2}{5}(x-6), \quad 2x-5y+23=0$ (o.e. with integer coefficients) M1, A1 4

Note

B1 for an expression for the gradient of AB . Does not need the $= -2.5$

1st M1 for use of the perpendicular gradient rule. Follow through their m

2nd M1 for the use of (6, 7) and their changed gradient to form an equation for l .

Can be awarded for $\frac{y-7}{x-6} = \frac{2}{5}$ o.e.

Alternative is to use (6, 7) in $y = mx + c$ to find a value for c .

Score when $c = \dots$ is reached.

A1 for a correct equation in the required form and must have “= 0” and integer coefficients

- (b) Using $x = 0$ in the answer to (a), $y = \frac{23}{5}$ or 4.6 M1, A1ft 2

Note

M1 for using $x = 0$ in their answer to part (a) e.g. $-5y + 23 = 0$

A1ft for $y = \frac{23}{5}$ provided that $x = 0$ clearly seen or $C(0, 4.6)$.

Follow through their equation in (a)

If $x = 0$, $y = 4.6$ are clearly seen but C is given as $(4.6, 0)$ apply ISW and award the mark.

This A mark requires a simplified fraction or an exact decimal

Accept their 4.6 marked on diagram next to C for M1A1ft

- (c) Area of triangle = $\frac{1}{2} \times 8 \times \frac{23}{5} = \frac{92}{5}$ (o.e) e.g. $\left(18\frac{2}{5}, 18.4, \frac{184}{10}\right)$ M1 A1 2

Note

M1 for $\frac{1}{2} \times 8 \times y_C$ so can follow through their y coordinate of C .

A1 for 18.4 (o.e.) but their y coordinate of C must be positive

Use of 2 triangles or trapezium and triangle

Award M1 when an expression for area of OCB only is seen

Determinant approach

Award M1 when an expression containing $\frac{1}{2} \times 8 \times y_C$ is seen

[8]

6. (a) $x = 2$: $y = 8 - 8 - 2 + 9 = 7$ (*) B1 1

Note

B1 there must be a clear attempt to substitute $x = 2$ leading to 7

e.g. $2^3 - 2 \times 2^2 - 2 + 9 = 7$

- (b) $\frac{dy}{dx} = 3x^2 - 4x - 1$ M1 A1

$x = 2$: $\frac{dy}{dx} = 12 - 8 - 1 (=3)$ A1ft

$y - 7 = 3(x - 2)$, $y = 3x + 1$ M1, A1 5

Note

1st M1 for an attempt to differentiate with at least one of the given terms fully correct.

1st A1 for a fully correct expression

2nd A1ft for sub. $x = 2$ in their $\frac{dy}{dx} = (\neq y)$ accept for a correct expression e.g. $3 \times (2)^2 - 4 \times 2 - 1$

2nd M1 for use of their “3” (provided it comes from their $\frac{dy}{dx} = (\neq y)$ and $x = 2$) to find equation of tangent. Alternative is to use (2, 7) in $y = mx + c$ to find a value for c . Award when $c = \dots$ is seen.

No attempted use of $\frac{dy}{dx}$ in (b) scores 0/5

(c) $m = -\frac{1}{3}$ (for $-\frac{1}{m}$ with their m) B1ft

$3x^2 - 4x - 1 = -\frac{1}{3}, 9x^2 - 12x - 2 = 0$ or $x^2 - \frac{4}{3}x - \frac{2}{9} = 0$ (o.e.) M1, A1

$\left(x = \frac{12 + \sqrt{144 + 72}}{18} \right) \left(\sqrt{216} = \sqrt{36} \sqrt{6} = 6\sqrt{6} \right)$ or $(3x - 2)^2$

$6 \rightarrow 3x = 2 \pm \sqrt{6}$ M1

$x = \frac{1}{3}(2 + \sqrt{6})$ (*) A1cso 5

Note

1st M1 for forming an equation from their $\frac{dy}{dx} = (\neq y)$ and their $-\frac{1}{m}$ (must be changed from m)

1st A1 for a correct 3TQ all terms on LHS (condone missing = 0)

2nd M1 for proceeding to $x = \dots$ or $3x = \dots$ by formula or completing the square for a 3TQ.

Not factorising. Condone $\pm$

2nd A1 for proceeding to given answer with no incorrect working seen. Can still have $\pm$.

ALT Verify (for M1A1M1A1)

1st M1 for attempting to square need ≥ 3 correct values in $\frac{4+6+4\sqrt{6}}{9}$, 1st A1 for $\frac{10+4\sqrt{6}}{9}$

2nd M1 Dependent on 1st M1 in this case for substituting in all terms of their $\frac{dy}{dx}$

2nd A1cso for cso with a full comment e.g. “the x co-ord of Q is ...”

[11]

7. (a) $y - 5 = -\frac{1}{2}(x - 2)$ or equivalent,

e.g. $\frac{y-5}{x-2} = -\frac{1}{2}$ $y = -\frac{1}{2}x + 6$ M1A1,A1cao 3

Note

M1 A1 The version in the scheme above can be written down directly (for 2 marks), and M1 A0 can be allowed if there is just one slip (sign or number).

If the 5 and 2 are the wrong way round the M mark can still be given if a correct formula (e.g. $y - y_1 = m(x - x_1)$) is seen, otherwise M0.

If (2, 5) is substituted into $y = mx + c$ to find c , the M mark is for attempting this and the 1st A mark is for $c = 6$.

Correct answer without working or from a sketch scores full marks.

(b) $x = -2 \Rightarrow y = -\frac{1}{2}(-2) + 6 = 7$ (therefore B lies on the line) B1 1
(or equivalent verification methods)

Note

A conclusion/comment is not required, except when the method used is to establish that the line through $(-2, 7)$ with gradient $-\frac{1}{2}$ has the same eqn. as found in part (a), or to establish that the line through $(-2, 7)$ and $(2, 5)$ has gradient $-\frac{1}{2}$. In these cases a comment 'same equation' or 'same gradient' or 'therefore on same line' is sufficient.

(c) $(AB^2 =) (2 - -2)^2 + (7 - 5)^2, = 16 + 4 = 20, AB = \sqrt{20} = 2\sqrt{5}$ M1, A1, A1 3

C is $(p, -\frac{1}{2}p + 6)$, so $AC^2 = (p - 2)^2 + \left(-\frac{1}{2}p + 6 - 5\right)^2$ M1

Note

M1 for attempting AB^2 or AB . Allow one slip (sign or number) inside a bracket, i.e. do not allow $(2 - -2)^2 - (7 - 5)^2$.

1st A1 for 20 (condone bracketing slips such as $-2^2 = 4$)

2nd A1 for $2\sqrt{5}$ or $k = 2$ (Ignore $\pm$ here).

(d) Therefore $25 = p^2 - 4p + 4 + \frac{1}{4}p^2 - p + 1$ M1

$25 = 1.25p^2 - 5p + 5$ or $100 = 5p^2 - 20p + 20$ (or better, RHS simplified A1 to 3 terms) Leading to: $0 = p^2 - 4p - 16(*)$ A1cso 4

Note

1st M1 for $(p-2)^2 + (\text{linear function of } p)^2$. The linear function may be unsimplified but must be equivalent to $ap + b$, $a \neq 0$, $b \neq 0$.

2nd M1 (dependent on 1st M) for forming an equation in p (using 25 or 5) and attempting (perhaps not very well) to multiply out both brackets.

1st A1 for collecting like p terms and having a correct expression.

2nd A1 for correct work leading to printed answer.

Alternative, using the result:

Solve the quadratic $(p = 2 \pm 2\sqrt{5})$ and use one or both of the two solutions to find the length of AC^2 or $C_1C_2^2$: e.g. $AC^2 = (2 + 2\sqrt{5} - 2)^2 + (5 - \sqrt{5} - 5)^2$ scores 1st M1, and 1st A1 if fully correct.

Finding the length of AC or AC^2 for both values of p , or finding C_1C_2 with some evidence of halving (or intending to halve) scores the 2nd M1.

Getting $AC = 5$ for both values of p , or showing $\frac{1}{2} C_1 C_2 = 5$ scores the 2nd A1 (cso).

[11]

8. (a) $\left(\frac{dy}{dx}\right) = -4 + 8x^{-2}$ (4 or $8x^{-2}$ for M1... sign can be wrong) M1A1

$x = 2 \Rightarrow m = -4 + 2 = -2$ M1

The first 4 marks could be earned in part (b)

$y = 9 - 8 - \frac{8}{2} = -3$ B1

Equation of tangent is: $y+3 = -2(x-2) \rightarrow y = 1 - 2x$ (*) M1 A1cso 6

Note

1st M1 for 4 or $8x^{-2}$ (ignore the signs).

1st A1 for both terms correct (including signs).

2nd M1 for substituting $x = 2$ into their $\frac{dy}{dx}$ (must be different from their y)

B1 for $yP = -3$, but not if clearly found from the given equation of the tangent.

3rd M1 for attempt to find the equation of tangent at P , follow through their m and yP .

Apply general principles for straight line equations (see end of scheme).

NO DIFFERENTIATION ATTEMPTED: Just assuming $m = -2$ at this stage is M0

2nd A1cso for correct work leading to printed answer (allow equivalents with $2x$, y , and 1 terms...

such as $2x + y - 1 = 0$).

(b) Gradient of normal = $\frac{1}{2}$

B1ft

Equation is: $\frac{y+3}{x-2} = \frac{1}{2}$ or better equivalent, e.g. $y = \frac{1}{2}x - 4$ M1A1 3

Note

B1ft for correct use of the perpendicular gradient rule. Follow through their m , but if $m \neq -2$ there must be clear evidence that the m is thought to be the gradient of the tangent.

M1 for an attempt to find normal at P using their changed gradient and their y_P .

Apply general principles for straight line equations (see end of scheme).

A1 for any correct form as specified above (correct answer only).

(c) $(A :) \frac{1}{2}, (B :) 8$

B1, B1

Area of triangle is: $\frac{1}{2} (x_B \pm x_A) \times y_P$ with values for all of x_B , x_A and y_P

M1

$\frac{1}{2} \left(8 - \frac{1}{2} \right) \times 3 = \frac{45}{4}$ or 11.25 A1 4

Note

1st B1 for $\frac{1}{2}$ and 2nd B1 for 8.

M1 for a full method for the area of triangle ABP . Follow through their x_A, x_B and their y_P , but the mark is to be awarded 'generously', condoning sign errors..

The final answer must be positive for A1, with negatives in the working condoned.

Determinant: Area = $\frac{1}{2} \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix} = \frac{1}{2} \begin{vmatrix} 2 & -3 & 1 \\ 0.5 & 0 & 1 \\ 8 & 0 & 1 \end{vmatrix} = \dots$ (Attempt

to multiply out required for M1)

Alternative: $AP = \sqrt{(2-0.5)^2 + (-3)^2}$, $BP = \sqrt{(2-8)^2 + (-3)^2}$ Area
 $= \frac{1}{2} AP \times BP = \dots$ M1

Intersections with y-axis instead of x-axis: Only the M mark is available B0 B0 M1 A0.

[13]

9. (a) $QR = \sqrt{(7-1)^2 + (0-3)^2}$ M1
 $= \sqrt{36+9}$ or $\sqrt{45}$ (condone $\pm$) A1
 $= 3\sqrt{5}$ or $a = 3$ ($\pm 3\sqrt{5}$ etc is A0) A1 3

M1 for attempting QR or QR^2 . May be implied by $6^2 + 3^2$

1st A1 for as printed or better. Must have square root. Condone $\pm$

(b) Gradient of QR (or l_1) $= \frac{3-0}{1-7}$ or $\frac{3}{-6}$, $= -\frac{1}{2}$ M1, A1
 Gradient of l_2 is $-\frac{1}{-\frac{1}{2}}$ or 2 M1
 Equation for l_2 is: $y-3 = 2(x-1)$ or $\frac{y-3}{x-1} = 2$ [or $y = 2x + 1$] M1A1ft 5

$y = 2x + 1$ with no working. Send to review.

1st M1 for attempting gradient of QR

1st A1 for -0.5 or $-\frac{1}{2}$ can be implied by gradient of $l_2 = 2$

2nd M1 for an attempt to use the perpendicular rule on their gradient of QR .

3rd M1 for attempting equation of a line using Q with their changed gradient.

2nd A1ft requires all 3 Ms but can ft their gradient of QR .

(c) P is $(0, 1)$ (allow " $x = 0, y = 1$ " but it must be clearly identifiable as P) B1 1

(d) $PQ = \sqrt{(1-x_p)^2 + (3-y_p)^2}$ M1
 $PQ = \sqrt{1^2 + 2^2} = \sqrt{5}$ A1
 Area of triangle is $\frac{1}{2} QR \times PQ = \frac{1}{2} 3\sqrt{5} \times \sqrt{5}$, $= \frac{15}{2}$ or 7.5 dM1, A1 4

Determinant Method

$$\text{e.g. } (0 + 0 + 7) - (1 + 21 + 0) \\ = -15 \text{ (o.e.)}$$

$$\text{Area} = \frac{1}{2} |-15| = 7.5$$

1st M1 for attempting PQ or PQ^2 follow through their coordinates of P

1st A1 for PQ as one of the given forms.

2nd dM1 for correct attempt at area of the triangle. Follow through their value of a and their PQ .
This M mark is dependent upon the first M mark

2nd A1 for 7.5 or some exact equivalent. Depends on both Ms.
Some working must be seen.

ALT

Use QS where S is (1, 0)

1st M1 for attempting area of $OPQS$ and QSR and OPR . Need all 3.

1st A1 for $OPQS = \frac{1}{2}(1 + 3) \times 1 = 2$, $QSR = 9$, $OPR = \frac{7}{2}$

2nd dM1 for $OPQS + QSR - OPR = \dots$ Follow through their values.

2nd A1 for 7.5

Determinant Method

M1 for attempt –at least one value in each bracket correct.

A1 if correct (± 15)

M1 for correct area formula

A1 for 7.5

MR

Misreading x -axis for y -axis for P . Do NOT use MR rule as this oversimplifies the question. They can only get M marks in (d) if they use PQ and QR .

Rules for quoting formula: For an M mark, if a correct formula is quoted and some correct substitutions seen then M1 can be awarded, if no values are correct then M0. If no correct formula is seen then M1 can only be scored for a fully correct expression.

[13]

10. (a) $m = \frac{4 - (-3)}{-6 - 8} \text{ or } \frac{-3 - 4}{8 - (-6)}, = \frac{7}{-14} \text{ or } \frac{-7}{14} \quad \left(= -\frac{1}{2} \right) \quad \text{M1, A1}$
- Equation: $y - 4 = -\frac{1}{2}(x - (-6)) \text{ or } y - (-3) = -\frac{1}{2}(x - 8) \quad \text{M1}$
- $x + 2y - 2 = 0$ (or equiv. with integer coefficients... must have '=' 0') A1 4
(e.g. $14y + 7x - 14 = 0$ and $14 - 7x - 14y = 0$ are acceptable)

1st M: Attempt to use $m = \frac{y_2 - y_1}{x_2 - x_1}$ (may be implicit in an equation of L).

2nd M: Attempting straight line equation in any form, e.g. $y - y_1 = m(x - x_1)$

$\frac{y - y_1}{x - x_1} = m$, with any value of m (except 0 or ∞) and either $(-6, 4)$ or $(8, -3)$.

N.B. It is also possible to use a different point which lies on the line, such as the midpoint of AB $(1, 0.5)$.

Alternatively, the 2nd M may be scored by using $y = mx + c$ with a numerical gradient and substituting $(-6, 4)$ or $(8, -3)$ to find the value of c .

Having coords the wrong way round, e.g. $y - (-6) = -\frac{1}{2}(x - 4)$,

loses the

2nd M mark unless a correct general formula is seen, e.g. $y - y_1 = m(x - x_1)$.

- (b) $(-6 - 8)^2 + (4 - (-3))^2 \quad \text{M1}$
- $14^2 + 7^2 \text{ or } (-14)^2 + 7^2 \text{ or } 14^2 + (-7)^2$ (M1 A1 may be implied by 245) A1
- $AB = \sqrt{14^2 + 7^2} \text{ or } \sqrt{7^2(2^2 + 1^2)} \text{ or } \sqrt{245} \quad \text{A1 cso} \quad 3$
- $7\sqrt{5}$

M: Attempting to use $(x_2 - x_1)^2 + (y_2 - y_1)^2$.

Missing bracket e.g. $-14^2 + 7^2$ implies M1 if no earlier version is seen.

$-14^2 + 7^2$ with no further work would be M1 A0.

$-14^2 + 7^2$ followed by 'recovery' can score full marks.

[7]

11. (a) $y = -\frac{3}{2}x(+4) \quad \text{Gradient} = -\frac{3}{2} \quad \text{M1A1} \quad 2$

M1 for an attempt to write $3x + 2y - 8 = 0$ in the form $y = mx + c$ or a full method that leads to $m =$, e.g. find 2 points, and

attempt gradient using $\frac{y_2 - y_1}{x_2 - x_1}$

e.g. finding $y = -1.5x + 4$ alone can score M1 (even if they

go on to say $m = 4$)

A1 for $m = -\frac{3}{2}$ (can ignore the +c) or $\frac{dy}{dx} = -\frac{3}{2}$

(b) $3x + 2 = -\frac{3}{2}x + 4$ $x = \dots, \frac{4}{9}$ M1, A1

$y = 3\left(\frac{4}{9}\right) + 2 = \frac{10}{3} \left(= 3\frac{1}{3}\right)$ A1 3

M1 for forming a suitable equation in one variable and attempting to solve leading to $x = \dots$ or $y =$

1stA1 for any extra correct value for x

2ndA1 for any exact correct value for y
(These 3 marks can be scored anywhere, they may treat (a) and (b) as a single part)

(c) Where $y = 1$, $l_1: x_A = -\frac{1}{3}$ $l_2: x_B = 2$ M: Attempt one of these M1A1

Area = $\frac{1}{2}(x_B - x_A)(y_P - 1)$ M1

$= \frac{1}{2} \times \frac{7}{3} \times \frac{7}{3} = \frac{49}{18} = 2\frac{13}{18}$ o.e. A1 4

1st M1 for attempting the x coordinate of A or B . One correct value seen scores M1.

1st A1 for $x_A = -\frac{1}{3}$ and $x_B = 2$

2nd M1 for a full method for the area of the triangle – follow through their x_A, x_B, y_P .

e.g. determinant approach $\frac{1}{2} \begin{vmatrix} 2 & -\frac{1}{3} & \frac{4}{9} & 2 \\ 1 & 1 & \frac{10}{3} & 1 \end{vmatrix} = \frac{1}{2} \left| 2 - \dots - \left(-\frac{1}{3} \dots\right) \right|$

2nd A1 for $\frac{49}{18}$ or an exact equivalent.

All accuracy marks require answers as single fractions or mixed numbers not necessary in lowest terms.

[9]

12. (a) $m = \frac{8-2}{11+1} (= \frac{1}{2})$ M1 A1
- $y-2 = \frac{1}{2}(x-11)$ or $y-8 = \frac{1}{2}(x-11)$ o.e. M1
- $y = \frac{1}{2}x + \frac{5}{2}$ accept exact equivalents e.g. $\frac{6}{12}$ A1c.a.o. 4
- 1st M1 for attempting $\frac{y_1-y_2}{x_1-x_2}$ must be y over x. No formula condone one sign slip but if formula is quoted then there must be some correct substitution.*
- 1st A1 for a fully correct expression, needn't be simplified.*
- 2nd M1 for attempting to find equation of l_1 .*
- (b) Gradient of $l_2 = -2$ M1
- Equation of l_2 $y-0 = -2(x-10)$ [$y = -2x + 20$] M1
- $\frac{1}{2}x + \frac{5}{2} = -2x + 20$ M1
- $x = 7$ and $y = 6$ depend on all 3 Ms A1, A1 5
- 1st M1 for using the perpendicular gradient rule*
- 2nd M1 for attempting to find equation of l_2 . Follow their gradient provided different.*
- 3rd M1 for forming a suitable equation to find S.*
- (c) $RS^2 = (10-7)^2 + (0-6)^2 (= 3^2 + 6^2)$ M1
- $RS = \sqrt{45} = 3\sqrt{5}$ A1c.s.o. 2
- M1 for expression for RS or RS^2 . Ft their S coordinates*

(d) $PQ = \sqrt{12^2 + 6^2} = 6\sqrt{5}$ or $\sqrt{180}$ or $PS = 4\sqrt{5}$ and $SQ = 2\sqrt{5}$ M1,A1

Area = $\frac{1}{2} PQ \times RS = \frac{1}{2} 6\sqrt{5} \times 3\sqrt{5}$ dM1

= 45

A1 c.a.o. 4

1st M1 for expression for PQ or PQ^2 . $PQ^2 = 12^2 + 6^2$

is M1 but $PQ = 12^2 + 6^2$ is M0

Allow one numerical slip.

2nd dM1 for a full, correct attempt at area of triangle.
Dependent on previous M1.

[15]

13. (a) $(y =)5 - 2 \times 3 = -1$

cso B1 1

(b) Gradient of perpendicular line is $\frac{1}{2}$

B1

$y - (-1) = -(x - 3)$

ft their $m \neq -2$ M1 A1ft

(or substituting $(3, -1)$ into $y = (\text{their } m)x + c$)

$x - 2y - 5 = 0$

A1 4

Ignore order but must be on one side and with integer coefficients.

May be any integer multiple of above.

[5]

14. (a) $y - (-4) = \frac{1}{3}(x - 9)$

M1A1

$x - 3y - 21 = 0$ or $3y - x + 21 = 0$

A1 3

(condone 3 terms equation: e.g. $x = 3y + 21$)

M1 for finding the equation of a straight line:

If using $y - y_1 = m(x - x_1)$ or equivalent, m must be $\frac{1}{3}$.

If using $y = mx + c$, m must be $\frac{1}{3}$ and c found.

First A1: unsimplified form

$[y - 9 = \frac{1}{3}(x + 4)$ M0, $y - 4 = \frac{1}{3}(x - 9)$ M1A0,

$y - 4 = \frac{1}{3}(x + 9)$ M0]

- (b) Equation of l_2 : $y = -2x$ B1
 Solve l_1 and l_2 simultaneously to find P : M1
 $x = 3, y = -6$ A1A1ft 4
M1: solving two linear equations to form linear equation in one variable
A1 for first coordinate if correct
A1 f.t.: For second coordinate correct for candidate after substituting in $y = -2x$
Watch $(-3, 6)$ [which usually scores M1A0A1 $\sqrt{\quad}$]
- (c) C: $(0, -7)$ or $OC = 7$ (may be on diagram) B1ft
 Area of triangle $OCP = \frac{1}{2} \times OC \times x_p = 10 \frac{1}{2}$ (must be exact) M1A1 3
B1 f.t.: Correct y value when $x = 0$ in candidate's equation in (a)
M1: For $\frac{1}{2} \times$ candidate's $OC \times$ |candidate's x co-ord in (b)|
SC: If x found when $y = 0$, allow M1 for finding area for their configuration.

[10]

15. (a) $p = 15, q = -3$ B1, B1 2
 Special case if B0 B0, allow M1 for method, e.g. $8 = \frac{1+p}{2}$
- (b) Gradient of line ADC = $-\frac{5}{7}$,
 gradient of perpendicular line = $-\frac{1}{\text{gradient ADC}} \left(\frac{7}{5} \right)$ B1, M1
 Equation of l : $y - 2 = \left(\frac{7}{5} \right)(x - 8)$ M1 A1ft
 $\Rightarrow 7x - 5y - 46 = 0$ (Allow rearrangements of this) A1cao 5
- (c) Substituting $y = 7$ and finding value for x , M1
 $x = \frac{81}{7}$ or $11 \frac{4}{7}$ A1 2

[9]

16. (a) $p = 15, q = -3$ B1 B1 2
- (b) Grad. of line ADC : $m = -\frac{5}{7}$, Grad. of perp. line $= -\frac{1}{m} \left(= \frac{7}{5} \right)$ B1, M1
 Equation of l : $y - 2 = \frac{7}{5}(x - 8)$ M1 A1ft
 $7x - 5y - 46 = 0$ (Allow rearrangements, e.g. $5y = 7x - 46$) A1 5
- (c) Substitute $y = 7$ into equation of l and find $x = \dots$ M1
 $\frac{81}{7}$ or $11\frac{4}{7}$ (or exact equiv.) A1 2

[9]

17. (a) Gradient of $AB = \frac{12-4}{11-(-1)} = \frac{2}{3}$ (or equiv.) M1 A1 2
- (b) Using $m_1 m_2 = -1$, gradient of $BC = -\frac{3}{2}$ B1ft
 Equation of BC : $y - 12 = -\frac{3}{2}(x - 11)$ M1 A1ft
 $3x + 2y - 57 = 0$ (Allow rearranged versions, e.g. $2y = 57 - 3x$) A1 4
- (c) D : $y = 0$ in equation of BC : $x = 19$ B1ft
 Coordinates of C : $(3, 24)$ B1, B1 3
- (d) $AC = \sqrt{("3"-(-1))^2 + ("24"-4)^2} = \sqrt{416} = 4\sqrt{26}$ (*) M1 A1 2

[11]

18. (a) $\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right) = \left(\frac{1+5}{2}, \frac{2+8}{2} \right) = (3, 5)$ M1 A1 2
- (b) Gradient $= \frac{y_2 - y_1}{x_2 - x_1} = \frac{8-2}{5-1}$ M1 A1
 $y - 2 = m(x - 1) \quad y = \frac{3}{2}x + \frac{1}{2}$ M1 A1 4
 Allow $y = \frac{3x+1}{2}$ or $y = \frac{1}{2}(3x + 1)$

[6]

19. (a) $\frac{5 - (-3)}{8 - 2} = \frac{4}{3}$ or equivalent M1 A1 2

(b) $M: \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right) (5, 1)$ M1 A1

Gradient of CM is $-\frac{3}{4}$ B1 ft

Equation of CM: $y - 1 = -\frac{3}{4}(x - 5)$ M1 A1 5

$(4y = -3x + 19)$

(Allow the M mark for an equation of a line through their mid-point, even with gradient $\frac{4}{3}$.)

(c) When $x = 4$, $y = \frac{7}{4}$ M1 A1 ft 2

(d) $r^2 = (4 - 2)^2 + \left(\frac{7}{4} + 3 \right)^2$ or $r^2 = (4 - 8)^2 + \left(\frac{7}{4} - 5 \right)^2$ M1 A1 ft

Other methods must be complete, up to r^2 .

Follow through only from their y value at the centre.

$= \sqrt{4 + \frac{361}{16}} = \sqrt{\frac{425}{16}} = \sqrt{\frac{25}{16}} \sqrt{17} = \frac{5\sqrt{17}}{4} (*)$ M1 A1 cso 4

The method mark requires both intermediate steps.

[13]

20. (a) $m = \frac{2 - 6}{12 - 4} \left(= -\frac{1}{2} \right)$ M1 A1

$y - 6 = (\text{their } m)(x - 4) \quad x + 2y = 16$ M1 A1 4

(b) $y = -4x$ B1 1

(c) $x + 2(-4x) = 16 \quad -7x = 16 \quad x = -\frac{16}{7}$ M1 A1

$y = \frac{64}{7}$ A1 ft

$A(4, 6), C\left(-\frac{16}{7}, \frac{64}{7}\right): \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right) \rightarrow \left(\frac{6}{7}, \frac{53}{7} \right)$ M1 A1 ft 5

[10]

21. (a) P: $x = 0$ $y = -2$ B1

Mid-point: $\left(\frac{(0+5)}{2}, \frac{(-2-3)}{2}\right) = \left(\frac{5}{2}, -\frac{5}{2}\right)$ M1 A1 ft

(b) Gradient of l_1 is $\frac{3}{2}$, so gradient of l_2 is $-\frac{2}{3}$ B1

$l_2: y - (-3) = -\frac{2}{3}(x - 5)$ M1 A1 ft

$2x + 3y = 1$ A1

(c) Solving: $3x - 2y = 4$

$2x + 3y = 1$ $x = \frac{14}{13}$ M1 A1

$y = \frac{-5}{13}$ M1 A1 ft

[11]

22. (a) Gradient of q is $\frac{7}{2}$ = gradient of p B1

Equation of p is $y - 2 = \frac{7}{2}(x + 3)$ or $7x - 2y = -25$ M1 A1 3

(b) p meets y -axis at $A\left(0, \frac{25}{2}\right)$ or q meets y -axis at $B(0, -7)$ M1

$AB = \frac{25}{2} - -7 = 19\frac{1}{2}$ A1 2

[5]

23. (a) Gradient of l_2 is $-\frac{1}{3}$ B1
- $y - 2 = -\frac{1}{3}(x - 6)$ $y = -\frac{1}{3}x + 4$ M1 A1 ft 3
- (b) $-\frac{1}{3}x + 4 = 3x - 6$ $x = 3$ M1 A1
- $y = 3$ A1 ft 3
- (c) $y = 0$; $l_1: x = 2$ $l_2: x = 12$ B1 B1 ft
- $(2, 0), (12, 0), (3, 3)$ Area of triangle = $\frac{1}{2}(10 \times 3) = 15$ M1 A1 4

[10]